

The global weight law: dependency structure and two closed bands

Program-status appendix (DAG), with the refereed band-closure corollaries

A. Longo Claude (CC/TA)*

July 10, 2026

Abstract

Target: the no-caveats global law — for every k -uniform tiling by unit regular polygons $\{3, 4, 6, 12\}$ (octagon analytic), $W(\Lambda) \leq B(k)$ with $B(1..3) = 5, 6, 7$ and $B(k) = 2k + 2\lfloor(k-1)/3\rfloor$ for $k \geq 4$, attained at every k . The proof is a case split over $(\text{hol}(\Lambda), \lambda_1)$; this appendix records the dependency structure (ten nodes), proves the two band closures that landed on 2026-07-10 — $\lambda_1 = 1$ and $\lambda_1 = \sqrt{3}$ (hexagon-bearing), each at $W(\Lambda) \leq 2k \leq B(k)$ for all $k \geq 4$, by exact layered-word climbs that bypass the honest local-walk constant $c_0 \approx 50$ — and documents two structural identifications: the width-2 inventory gap 3.1(d) equals the band program's E4-A' (one finite check, two ledgers, all 378 catalogued $\lambda_1 = 2$ tilings), and the snub band is a two-letter layered word family whose closure follows the same pattern. No enumeration data enters any proof; the catalogue appears only as corroboration.

1 Target and case split

$W(\Lambda)$ is the least possible maximum step-weight of a basis of the period lattice (12-direction alphabet). The split: $\text{hol}(\Lambda) \in \{8, 12\}$ (square/hexagonal) is the rigid branch; $\text{hol} \leq 4$ splits by λ_1 . Status of every cell (2026-07-10):

*Program design and drafting: CC, also acting as TA (A. Longo's authorization, 2026-07-10). Corollaries C1/C2 hardened by two adversarial referee subagents (no fatal findings; both bounds came back sharper). Sources of truth: `docs/WEIGHT_PROOF_DAG.md` and `resources/research/th10-D3-consolidation-2026-07-10.md`.

cell	bound	status	carrier
$k \leq 3$, all cells	5, 6, 7 exact	CLOSED	small- k appendix (refereed)
rigid, $4 \leq k \leq K^*$	census per k	OPEN (compute)	D8: shell engine
rigid, $k > K^*$	$c\sqrt{k} + d$	OPEN	D9 (rate machinery)
$\lambda_1 = 1$	$W \leq 2k$	CLOSED (Corollary 2.3)	D3, this appendix
$\lambda_1 \in (1, \sqrt{3})$	—	OPEN (measured empty)	D4: extended T2
$\lambda_1 = \sqrt{3}$, hex-bearing	$W \leq 2k$	CLOSED (Corollary 2.4)	D3, this appendix
$\lambda_1 = \sqrt{3}$, $\{3, 4\}$ -only	—	OPEN (measured empty)	D4: extended T2
$\lambda_1 = 2$ (all hex-bearing)	$B(k)$ exact	MOD 3.1(D)	width-2 IP ($k \geq 4$ appendix)
$\lambda_1 = 2 \cos 15^\circ$ (snub)	$(1 + \sqrt{3})k$ crude	OPEN (route fixed)	D6: row-word classification
other shells in $(\sqrt{3}, 1 + \sqrt{2})$	—	OPEN	shell census (engine incr. 2)
$[1 + \sqrt{2}, T_0)$ wide	$\sqrt{3}H \leq \sim 4.2k$	OPEN	D7: E5 + fine rate
$\geq T_0$	crossing constants	MOD T_0	D5 sharpening
attainment, all k	$B(k)$ attained	CLOSED	construction (Thm B) + sma

Critical path: D1 (slab engine) $\rightarrow$ D6 (snub) $\rightarrow$ D10 (assembly). The engine's increment 1a (2026-07-10) reproduces the width-2 $\{T, S, H\}$ slab world mechanically; increment 1b (exhaustive fronts) closes D2 = 3.1(d) = E4-A'; increment 2 (rotated frames) serves D6 and the shell census.

2 The shared lemmas and the two closed bands

Throughout: u a shortest period, the tiling drawn on the λ_1 -cylinder, “level” = a horizontal line through a vertex, heights measured along the cylinder axis.

Lemma 2.1 (one vertex class per level, $\lambda_1 < 2$). *Distinct vertices are at distance ≥ 1 (thesis lem:wallpaper, $\delta_0 = 1$). Two vertex classes on one level of the λ_1 -cylinder have arc distances δ and $\lambda_1 - \delta$; both ≥ 1 forces $\lambda_1 \geq 2$. So for $\lambda_1 < 2$ each level carries exactly one class.*

Lemma 2.2 (canonical pair). *A shortest period u is primitive; any horizontal period is an integer multiple of u ; hence $\ker(\text{height}) \cap \Lambda = \mathbb{Z}u$ and $\text{height}(\Lambda) = h_0\mathbb{Z}$ with $h_0 > 0$ minimal. Any $c' \in \Lambda$ of height h_0 makes (u, c') a basis.*

Proof. $u = mw$, $m \geq 2$ contradicts minimality; $v = tu$, $t \notin \mathbb{Z}$ leaves the shorter horizontal period $\{t\}u$. For the basis: $x - (\text{height}(x)/h_0)c'$ has height 0, hence lies in $\mathbb{Z}u$. $\square$

Corollary 2.3 ($\lambda_1 = 1$ closes at $W \leq 2k$). *Every $\{3, 4, 6, 12\}$ -tiling with $\lambda_1 = 1$ satisfies $W(\Lambda) \leq 2k \leq B(k)$ for all $k \geq 4$ (at $k = 4$: band $\max \leq 8 < 10 = B(4)$, margin).*

Proof. By the Width Lemma only triangles and squares fit, and by the layered classification (E2, with the E-12 repair; blocker ledgered in Section 5) the tiling is a stack of p rings: square rings (height 1) and triangle rings (height $\sqrt{3}/2$), whose interfaces carry the vertex set $\delta + \mathbb{Z}$ — one class per interface (also Lemma 2.1), $|V(Q)| = p$, and $p \leq 2k$ (a primitive word's dihedral symmetry group has order ≤ 2 , so $k = \#\text{orbits} \geq \lceil p/2 \rceil$; an inessential vertical glide fixes interface indices, an essential one is dead for primitive words). Climb: every interface vertex of a square ring carries a vertical edge to the unique upper class; every base vertex $\delta + n$ of a triangle ring is a corner of the up-triangle on $[\delta + n, \delta + n + 1]$, whose 60° -edge lands at $\delta + n + \frac{1}{2}$ — the unique upper class. One edge per ring, no connectors.

Endpoint: p rings span exactly one Λ -period of height H ; $\text{height}(\Lambda) = H\mathbb{Z}$ (Lemma 2.2); the landing vertex lies on the target level, whose unique class forces it to be $v + c + mu =: v + c'$ with $\text{height}(c') = H$ minimal, so (u, c') is a basis. $\text{wt}(u) = 1$ (u is literally an interface edge) and $\text{wt}(c') \leq p \leq 2k$. $\square$

Corollary 2.4 ($\lambda_1 = \sqrt{3}$, hexagon-bearing, closes at $W \leq 2k$). *Every hexagon-bearing $\{3, 4, 6\}$ -tiling with $\lambda_1 = \sqrt{3}$ satisfies $W(\Lambda) \leq 2k \leq B(k)$ — in fact for all $k \geq 1$, matching the measured band maximum exactly.*

Proof. By the ring classification (E3, patched: Chord-Lemma form of the hexagon-orientation lemma, crystallographic exclusion, level-gap-word recovery; sharpened count $|V(Q)| = 2h + t \leq 2k$, now machine-checked on all 55 in-band catalogue tilings), the tiling is a word in HEX blocks (height $3/2$, level gaps $\frac{1}{2}$ then 1) and TRI blocks (height $\frac{1}{2}$, one gap). Each level gap is crossed by exactly one edge: the $\frac{1}{2}$ -gaps by one of the two boundary-zigzag edges at the valley (direction 30° or 150° , climb $\sin 30^\circ = \frac{1}{2}$), the 1-gaps by the vertical hexagon edge whose lower endpoint the zigzag edge just reached (corner table of the orientation lemma). The walk is monotone and crosses each of the $|V|$ gaps once per period: steps $= |V| \leq 2k$. Endpoint and basis by Lemma 2.2; $\text{wt}(u) = 2$ ($|u| = \sqrt{3} > 1$ gives ≥ 2 ; $e^{i30^\circ} + e^{-i30^\circ} = \sqrt{3}$ gives ≤ 2). Hence $W(\Lambda) \leq \max(2, 2k) = 2k$. $\square$

Remark 2.5 (why these are new). Both bands previously carried $s^* \leq \frac{2}{\sqrt{3}}H + c_0$ resp. $\frac{4}{3}H + c_0$ with the honest $c_0 \approx 50$ (adversarial review of 2026-07-03), useless below $k \approx 40$. The word climbs above have additive constant 0: the layered structure is not just a counting device but an exact routing device. The same pattern is the planned route for the snub band (Section 3).

3 Identifications and the snub re-scope

3.1(d) = E4-A' = node D2. The width-2 appendix's only conditionality (exclusion of vertical-edge triangles and 30° -tilted squares at circumference 2) is the same obligation the band program flags as E4-A'; the corrected census shows every $\lambda_1 = 2$ catalogue tiling (378) is hexagon-bearing and S/T/X-layered, so one finite check discharges: Theorems A/C's condition, the E4-A classification, and the $\lambda_1 = 2$ row of Section 1. The slab engine's exhaustive-front increment is the designated closer; its reachable-closure increment already reproduces the inventory.

The snub band re-scope (honesty entry). The hoped-for shortcut — “every weak climbing step (best up-edge $\sin 45^\circ$) lands at a vertex with a strong ($\sin 75^\circ$) up-edge” — is FALSE: the machine sweep finds all four weak-landing direction sets contain an adjacent-square (domino) corner, and the catalogue realizes 829 such vertices across 181 snub-band tilings. Two traps were dodged and recorded: the domino does *not* overlap its u -translate (support width is not an overlap criterion — the review's E-4 lesson), and infinite square strips are not excluded by lattice arithmetic (they force a legal 45° -direction period). The surviving route is visible in the catalogue's own naming (34-5d2_5f etc.): the band is a two-letter layered word family; classify the rows (E2/E3-style), then climb the word exactly as in Corollaries 2.3 and 2.4. The engine's rotated-frame increment (15° -offset) is the discovery and completeness tool.

4 Referee round

Two adversarial referee subagents attacked Corollaries 2.3 and 2.4 (2026-07-10). No fatal findings; both bounds came back sharper: C1 from $2k+2$ to $2k$ (orbit-count argument) with $\text{wt}(u) = 1$ needing no lemma at all; C2's count upgraded from data-observed to machine-checked (the V5 assert tightened to the sharpened $|V| \leq 2k$, 55/55). Gaps repaired in place: Lemmas 2.1 and 2.2 written out (they had been consumed silently); the endpoint lemma written (the exact failure mode the 2026-07-03 review flagged as G-4 recurred and is now dead); the per-block climb gloss replaced by the gap-crossing invariant; the statement re-anchored to $W(\Lambda)$ (the quantity the enumerator's pair pool consumes) rather than the generator radius s^* .

5 Status ledger

- **CLOSED**: Lemmas 2.1 and 2.2 and Corollaries 2.3 and 2.4 (refereed); the engine increment 1a regression; the identifications of Section 3.
- **MOD E2-v2**: Corollary 2.3 walks through E2's ring decomposition, whose Step-4 circularity repair (the "three-lemma band-propagation restructure", reviewer-supplied in outline) is not yet integrated on disk. Writing E2-v2 is the one blocker between the D3 node and DONE.
- **OPEN** (by node): D2/3.1(d) (engine increment 1b: exhaustive fronts with incremental simplicity pruning); D4 extended T2 (three empty-band claims); D5 sharp crossing constants; D6 snub row classification; D7 wide bands; D8/D9 rigid branch; the per-band shell census; D10 assembly.
- Solver-side (independent of this program): the $k \leq 3$ re-run at the proven small- k config — the one item the thesis's $k = 3$ completeness claim still needs.